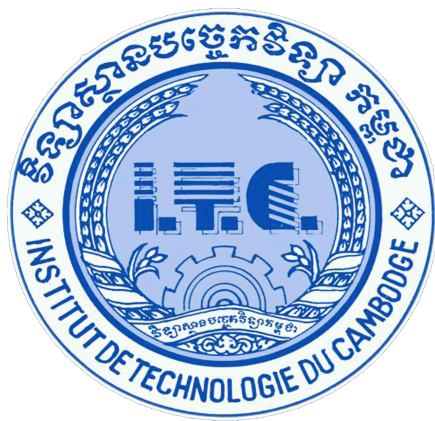




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Course: Networks System Design

Instructor: KUY Movsun

Assignment: Lab9(Network Layer Data Plane)

Due Date: December 30, 2025 (12:00 AM)

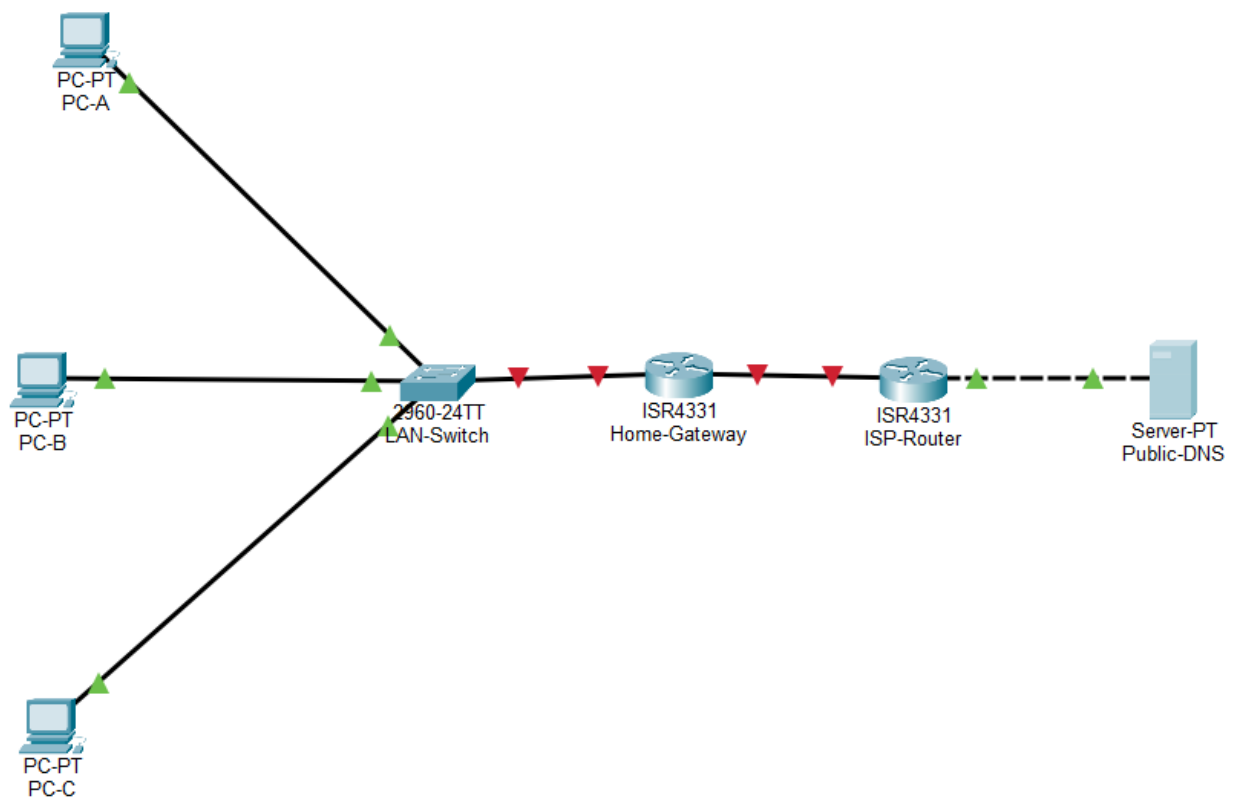
Part 1: Topology & Addressing

Device Table

Device Name	Model	Interface	Description
ISP-Router	ISR 4331	Gig0/0/0	WAN Link (to Home-Gateway)
		Gig0/0/1	Internet Link (to Server)
Home-Gateway	ISR 4331	Gig0/0/0	WAN Uplink (to ISP)
		Gig0/0/1	LAN Gateway (to Switch)
LAN-Switch	2960	Any	Layer 2 Switch
Public-DNS	Server-PT	NIC	Simulated Internet Server (8.8.8.8)

Cabling Guide

- ISP-Router (G0/0/0) ↔ Home-Gateway (G0/0/0): Copper Crossover (or Straight-through if Auto-MDIX)
- ISP-Router (G0/0/1) ↔ Public-DNS: Copper Crossover
- Home-Gateway (G0/0/1) ↔ LAN-Switch (G0/1): Copper Straight-through
- LAN-Switch ↔ PC-A, PC-B, PC-C: Copper Straight-through



Part 2: Pre-Lab Setup (Instructor Configuration)

Configure Public-DNS Server

1. Go to **Public-DNS > Desktop > IP Configuration**
2. Set:
 - IP Address: **8.8.8.8**
 - Subnet Mask: **255.255.255.0**
 - Default Gateway: **8.8.8.1**

Configure ISP-Router (CLI)

```
Router> enable
Router# configure terminal
Router(config)# hostname ISP-Router

! Connection to Server
Router(config)# interface g0/0/1
Router(config-if)# ip address 8.8.8.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

! Connection to Home-Gateway
Router(config)# interface g0/0/0
Router(config-if)# ip address 203.0.113.1 255.255.255.252
```

```
Router(config-if)# no shutdown
Router(config-if)# end
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ISP-Router
ISP-Router(config)# interface g0/0/1
ISP-Router(config-if)#ip address 8.8.8.1 255.255.255.0
ISP-Router(config-if)#no shutdown

ISP-Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
exit
ISP-Router(config)#exit
ISP-Router#
%SYS-5-CONFIG_I: Configured from console by console

ISP-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ISP-Router(config)#interface g0/0/0
ISP-Router(config-if)#ip address 203.0.113.1 255.255.255.252
ISP-Router(config-if)#no shutdown

ISP-Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

ISP-Router(config-if)#end
ISP-Router#
%SYS-5-CONFIG_I: Configured from console by console

ISP-Router#
```

Task 3: Implementing NAT (Network Address Translation)

Scenario

Convert the LAN to a private network (**192.168.1.0/24**) and configure NAT Overload (PAT) so internal hosts can access the Internet.

Step 3A: Configure WAN Link & Default Route

```
Home-Gateway(config)# interface g0/0/0
Home-Gateway(config-if)# description WAN-Link
Home-Gateway(config-if)# ip address 203.0.113.2 255.255.255.252
Home-Gateway(config-if)# no shutdown
Home-Gateway(config-if)# exit

! Default route to ISP
Home-Gateway(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

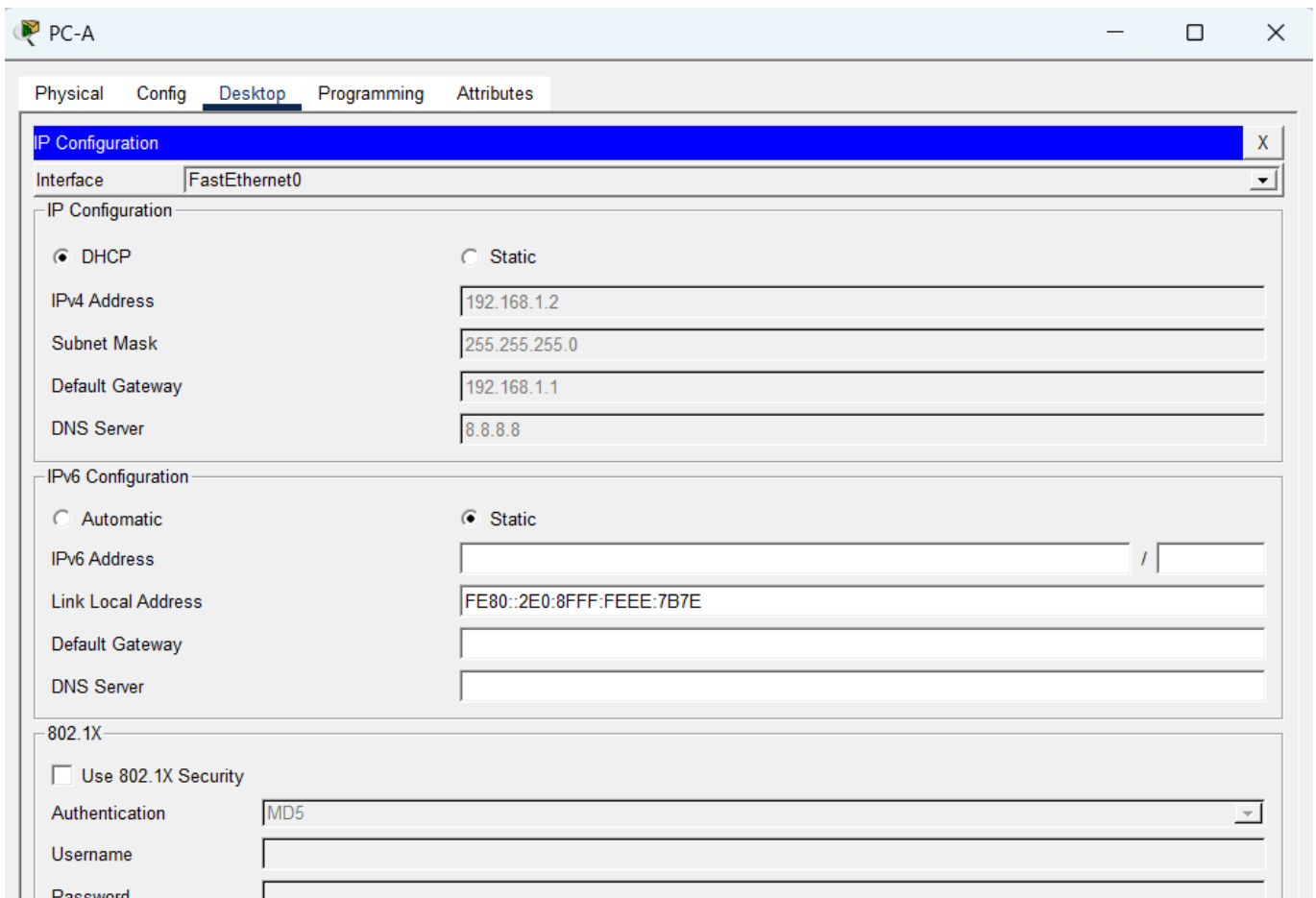
Step 3B: Re-configure LAN as Private Network

```
! Change LAN Gateway IP
Home-Gateway(config)# interface g0/0/1
Home-Gateway(config-if)# ip address 192.168.1.1 255.255.255.0
Home-Gateway(config-if)# exit

! Update DHCP Pool
Home-Gateway(config)# no ip dhcp pool LAN_POOL
Home-Gateway(config)# ip dhcp pool PRIVATE_LAN
Home-Gateway(dhcp-config)# network 192.168.1.0 255.255.255.0
Home-Gateway(dhcp-config)# default-router 192.168.1.1
Home-Gateway(dhcp-config)# dns-server 8.8.8.8
Home-Gateway(dhcp-config)# exit
```

Step 3C: Force PCs to renew IP

On PC-A, PC-B, and PC-C: Open the IP Configuration window, switch to Static, wait a second, then switch back to DHCP. Verify they now have an address like 192.168.1.2.



Step 3D: Configure NAT Overload (PAT)

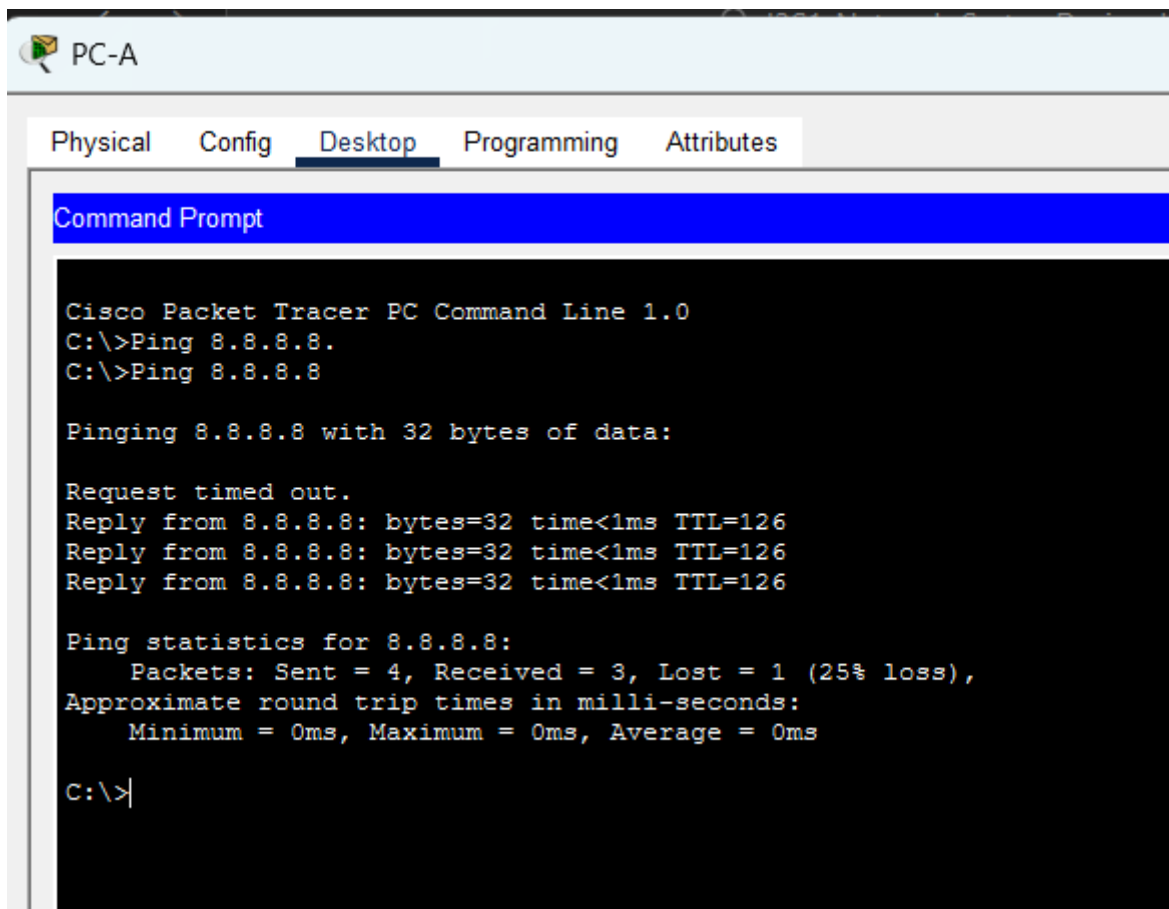
```
! Define Inside and Outside interfaces
Home-Gateway(config)# interface g0/0/1
Home-Gateway(config-if)# ip nat inside
Home-Gateway(config-if)# exit
```

```
Home-Gateway(config)# interface g0/0/0
Home-Gateway(config-if)# ip nat outside
Home-Gateway(config-if)# exit

! Create ACL for LAN traffic
Home-Gateway(config)# access-list 1 permit 192.168.1.0 0.0.0.255

! Apply NAT Overload
Home-Gateway(config)# ip nat inside source list 1 interface g0/0/0 overload
```

- Verification: Open Command Prompt on PC-A. Ping 8.8.8.8. If configured correctly, the ping will succeed.



- Check the router table: show ip nat translations.

```
Home-Gateway#
%SYS-5-CONFIG_I: Configured from console by console

Home-Gateway#show ip nat translation
Pro Inside global      Inside local           Outside local          Outside global
udp 203.0.113.2:1025    192.168.1.2:1025      8.8.8.8:53            8.8.8.8:53

Home-Gateway#
```

```
! Enable IPv6 routing
Home-Gateway(config)# ipv6 unicast-routing
```

```
! Configure IPv6 on LAN interface
Home-Gateway(config)# interface g0/0/1
Home-Gateway(config-if)# ipv6 address 2001:DB8:ACAD:1::1/64
Home-Gateway(config-if)# no shutdown
Home-Gateway(config-if)# exit
```

- Verification: On PC-C > IP Configuration, look at the "IPv6 Configuration" section. Ensure it is set to Auto Config. It should automatically generate an IPv6 address.

The screenshot shows the 'PC-C' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing the 'FastEthernet0' interface. Under 'IP Configuration', the 'DHCP' radio button is selected, and a message states 'DHCP request successful.' The fields for IPv4 Address, Subnet Mask, Default Gateway, and DNS Server are populated with values. The 'IPv6 Configuration' section is also expanded, showing the 'Automatic' radio button selected, with a message 'IPv6 request successful.' The IPv6 Address field is populated with '2001:DB8:ACAD:1:230:F2FF:FE70:7AE3' and a prefix of '/64'. The Link Local Address, Default Gateway, and DNS Server fields are also populated. The '802.1X' section is expanded, showing the 'Use 802.1X Security' checkbox unchecked, and the Authentication dropdown set to 'MD5'. The Username and Password fields are empty.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☒ DHCP	☐ Static
DHCP request successful.	
IPv4 Address	192.168.1.4
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8
IPv6 Configuration	
☒ Automatic	☐ Static
IPv6 request successful.	
IPv6 Address	2001:DB8:ACAD:1:230:F2FF:FE70:7AE3 / 64
Link Local Address	FE80::230:F2FF:FE70:7AE3
Default Gateway	FE80::260:2FFF:FE44:CA02
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	